

casino_clean_data

May 13, 2026

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[16]: import pdfplumber
import pandas as pd
import re
import os

# STEP 1: Set base folder path
BASE_FOLDER = r"C:\Users\abidh\Documents"
OUTPUT_FOLDER = r"C:\Users\abidh\Documents"

FOLDERS = {
    "Monthly Gross Revenue Report 2022": ("casino_revenue", 2022),
    "Monthly Gross Revenue Report 2023": ("casino_revenue", 2023),
    "Monthly Gross Revenue Report 2024": ("casino_revenue", 2024),
    "Monthly Gross Revenue Report 2025": ("casino_revenue", 2025),
    "Monthly igaming tax return 2022": ("igaming", 2022),
    "Monthly igaming tax return 2023": ("igaming", 2023),
    "Monthly igaming tax return 2024": ("igaming", 2024),
    "Monthly igaming tax return 2025": ("igaming", 2025),
    "Monthly sports wagering tax return 2022": ("sports_wagering", 2022),
    "Monthly sports wagering tax return 2023": ("sports_wagering", 2023),
    "Monthly sports wagering tax return 2024": ("sports_wagering", 2024),
    "Monthly sports wagering tax return 2025": ("sports_wagering", 2025),
}

# STEP 2: Number extractor - handles split numbers like "$ 1 3,437,426"
def extract_first_dollar(line):
    """Extracts the first dollar amount from a line, handles split formatting.
    ↪ """
    # Match $ followed by digits/spaces/commas - joins split numbers like "$ 1_
    ↪ 3,437,426"
    parts = re.findall(r'\$\s*([\d\s,]+)', line)
    if parts:
        combined = re.sub(r'\s+', '', parts[0])
        try:
            val = float(combined)
            return val if val > 0 else None
        except:
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        return None
    # Fallback: match plain numbers (no $ sign) for sports wagering lines
    match = re.search(r'\b(\d{1,3}(?:,\d{3})+|\d{5,})\b', line)
    if match:
        try:
            return float(match.group().replace(',', ''))
        except:
            return None
    return None

def extract_last_dollar(line):
    """Extracts the LAST dollar amount from a line - useful for YTD fields."""
    parts = re.findall(r'\$\s*([\d\s,]+)', line)
    if parts:
        combined = re.sub(r'[\s,]', '', parts[-1])
        try:
            val = float(combined)
            return val if val > 0 else None
        except:
            return None
    return None

def clean_number(text):
    """Clean a raw number string including parentheses for negatives."""
    if not text:
        return None
    cleaned = re.sub(r'[\$, \s]', '', str(text))
    cleaned = cleaned.replace('(', '-').replace(')', '')
    try:
        return float(cleaned)
    except:
        return None

# STEP 3: Get month/year from text
def get_month_year(text, default_year):
    match = re.search(r'FOR THE MONTH OF\s+([A-Z]+)[,.\s]+(\d{4})', text, re.
↳ IGNORECASE)
    if match:
        return match.group(1).capitalize(), int(match.group(2))
    return None, default_year

# STEP 4: Get casino name
def get_casino_name(text):
    match = re.search(r'^(.+?)\n(?:MONTHLY|FOR THE MONTH)', text, re.MULTILINE)
    if match:
        name = match.group(1).strip()

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        name = re.sub(r',?\s*(LLC|LP|INC|CORP|LTD).*$', '', name, flags=re.
↳IGNORECASE)
        name = re.sub(r'\s+dba\s*$', '', name, flags=re.IGNORECASE)
        if len(name) > 3:
            return name.strip()
        match = re.search(
            r'\n([A-Z][A-Z\s&\'\.]+(?:
↳CASINO|RESORT|HOTEL|GAMING|RACEWAY|RACETRACK|PARK)[A-Z\s&\'\.]*)\n',
            text
        )
        if match:
            return match.group(1).strip()
        return "Unknown"

# STEP 5: Extract CASINO REVENUE
def extract_casino_revenue(lines, text, filename, year):
    data = {
        'data_type': 'casino_revenue',
        'source_file': filename,
        'casino': get_casino_name(text),
        'table_games_win': None,
        'poker_win': None,
        'slot_win': None,
        'total_casino_win': None,
        'ytd_gross_revenue': None,
    }
    data['month'], data['year'] = get_month_year(text, year)

    for line in lines:
        lu = line.upper()
        if re.match(r'^1\s+TABLE AND OTHER GAMES', lu):
            data['table_games_win'] = extract_first_dollar(line)
        elif re.match(r'^2\s+POKER', lu):
            data['poker_win'] = extract_first_dollar(line)
        elif re.match(r'^[34]\s+SLOT MACHINE', lu):
            data['slot_win'] = extract_first_dollar(line)
        elif re.match(r'^[345]\s+TOTAL CASINO WIN', lu):
            data['total_casino_win'] = extract_first_dollar(line)
        elif re.match(r'^10\s+YEAR-TO-DATE GROSS REVENUE', lu):
            data['ytd_gross_revenue'] = extract_last_dollar(line)
        elif re.match(r'^[789]\s+YEAR-TO-DATE', lu) and 'GROSS REVENUE' in lu:
            val = extract_last_dollar(line)
            if val:
                data['ytd_gross_revenue'] = val

    return data

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# STEP 6: Extract IGAMING
def extract_igaming(lines, text, filename, year):
    if 'SKIN DETAIL' in text.upper():
        return None

    data = {
        'data_type': 'igaming',
        'source_file': filename,
        'casino': get_casino_name(text),
        'monthly_total_igaming_win': None,
        'ytd_total_igaming_win': None,
        'ytd_igaming_gross_revenue': None,
    }
    data['month'], data['year'] = get_month_year(text, year)

    for line in lines:
        lu = line.upper()
        if re.match(r'^3\s+TOTAL', lu):
            data['monthly_total_igaming_win'] = extract_first_dollar(line)
        elif re.match(r'^6\s+YEAR-TO-DATE TOTAL', lu):
            data['ytd_total_igaming_win'] = extract_last_dollar(line)
        elif re.match(r'^8\s+YEAR-TO-DATE INTERNET GAMING GROSS', lu):
            data['ytd_igaming_gross_revenue'] = extract_last_dollar(line)

    return data

# STEP 7: Extract SPORTS WAGERING
def extract_sports_wagering(lines, text, filename, year):
    data = {
        'data_type': 'sports_wagering',
        'source_file': filename,
        'casino': get_casino_name(text),
        'monthly_retail_sports_win': None,
        'monthly_internet_sports_win': None,
        'total_sports_tax_payment': None,
    }
    data['month'], data['year'] = get_month_year(text, year)

    for line in lines:
        lu = line.upper()
        if re.match(r'^3\s+MONTHLY RETAIL SPORTS', lu):
            val = extract_first_dollar(line)
            if val is None:
                match = re.search(r'[\d,]+\$', line.strip())
                if match:
                    val = clean_number(match.group())
            data['monthly_retail_sports_win'] = val

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        elif re.match(r'^16\s+MONTHLY (INTERNET|ONLINE) SPORTS', lu):
            val = extract_first_dollar(line)
            if val is None:
                match = re.search(r'[\d,]+\$', line.strip())
                if match:
                    val = clean_number(match.group())
            data['monthly_internet_sports_win'] = val

        elif re.match(r'^26\s+TOTAL SPORTS WAGERING TAX', lu):
            data['total_sports_tax_payment'] = extract_last_dollar(line)

    return data

# STEP 8: Route each page to the correct extractor
def extract_page(page, filename, data_type, year):
    text = page.extract_text()
    if not text:
        return None

    lines = text.split('\n')
    tu = text.upper()

    if data_type == 'casino_revenue':
        if 'MONTHLY GROSS REVENUE REPORT' not in tu:
            return None
        return extract_casino_revenue(lines, text, filename, year)

    elif data_type == 'igaming':
        if 'MONTHLY INTERNET' not in tu:
            return None
        if 'SKIN DETAIL' in tu:
            return None
        return extract_igaming(lines, text, filename, year)

    elif data_type == 'sports_wagering':
        if 'MONTHLY SPORTS WAGERING TAX RETURN' not in tu:
            return None
        return extract_sports_wagering(lines, text, filename, year)

    return None

# STEP 9: Loop through all 12 folders
all_records = []
total_files = 0
total_pages = 0

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for folder_name, (data_type, year) in FOLDERS.items():
    folder_path = os.path.join(BASE_FOLDER, folder_name)

    if not os.path.exists(folder_path):
        print(f"WARNING: Folder not found - {folder_path}")
        continue

    pdf_files = [f for f in os.listdir(folder_path) if f.lower().endswith('.
pdf')]
    print(f" {folder_name} - {len(pdf_files)} PDFs found")
    total_files += len(pdf_files)

    for filename in pdf_files:
        filepath = os.path.join(folder_path, filename)
        try:
            with pdfplumber.open(filepath) as pdf:
                for page in pdf.pages:
                    record = extract_page(page, filename, data_type, year)
                    if record and record.get('casino') not in ["Unknown", None]:
                        all_records.append(record)
                        total_pages += 1
        except Exception as e:
            print(f" ERROR: {filename} - {e}")

# STEP 10: Save to CSVs
df = pd.DataFrame(all_records)

df['date'] = pd.to_datetime(
    df['month'] + ' ' + df['year'].astype(str),
    format='%B %Y', errors='coerce'
)
df = df.sort_values(['data_type', 'casino', 'date']).reset_index(drop=True)

# Master CSV
master_path = os.path.join(OUTPUT_FOLDER, "all_casino_data.csv")
df.to_csv(master_path, index=False)

# Separate CSVs
for dtype in df['data_type'].unique():
    subset = df[df['data_type'] == dtype]
    out_path = os.path.join(OUTPUT_FOLDER, f"{dtype}_clean.csv")
    subset.to_csv(out_path, index=False)
    print(f"Saved {len(subset)} records → {dtype}_clean.csv")

print(f"\nDone!")
print(f"Total PDFs processed    : {total_files}")
print(f"Total records extracted: {total_pages}")

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print(f"\nBreakdown:")
print(df['data_type'].value_counts())

# Preview Ocean Casino across all 3 data types
print("\nPreview - Casino Revenue (Ocean Casino):")
casino_df = df[df['data_type'] == 'casino_revenue']
ocean = casino_df[casino_df['casino'].str.contains('OCEAN', case=False, na=False)]
print(ocean[['casino', 'month', 'year', 'table_games_win', 'slot_win', 'total_casino_win']].head(10).to_string())

print("\nPreview - iGaming (Ocean Casino):")
ig_df = df[df['data_type'] == 'igaming']
ocean_ig = ig_df[ig_df['casino'].str.contains('OCEAN', case=False, na=False)]
print(ocean_ig[['casino', 'month', 'year', 'monthly_total_igaming_win', 'ytd_igaming_gross_revenue']].head(10).to_string())

print("\nPreview - Sports Wagering (Ocean Casino):")
sw_df = df[df['data_type'] == 'sports_wagering']
ocean_sw = sw_df[sw_df['casino'].str.contains('OCEAN', case=False, na=False)]
print(ocean_sw[['casino', 'month', 'year', 'monthly_retail_sports_win', 'monthly_internet_sports_win']].head(10).to_string())

```

Monthly Gross Revenue Report 2022 - 12 PDFs found
 Monthly Gross Revenue Report 2023 - 12 PDFs found
 Monthly Gross Revenue Report 2024 - 12 PDFs found
 Monthly Gross Revenue Report 2025 - 12 PDFs found
 Monthly igaming tax return 2022 - 12 PDFs found
 Monthly igaming tax return 2023 - 12 PDFs found
 Monthly igaming tax return 2024 - 12 PDFs found
 Monthly igaming tax return 2025 - 12 PDFs found
 Monthly sports wagering tax return 2022 - 12 PDFs found
 Monthly sports wagering tax return 2023 - 12 PDFs found
 Monthly sports wagering tax return 2024 - 12 PDFs found
 Monthly sports wagering tax return 2025 - 12 PDFs found
 Saved 432 records → casino_revenue_clean.csv
 Saved 394 records → igaming_clean.csv
 Saved 708 records → sports_wagering_clean.csv

Done!
 Total PDFs processed : 144
 Total records extracted: 1534

Breakdown:
 data_type
 sports_wagering 708
 casino_revenue 432

```
igaming          394
Name: count, dtype: int64
```

Preview - Casino Revenue (Ocean Casino):

	casino	month	year	table_games_win	slot_win
total_casino_win					
252	OCEAN CASINO RESORT	February	2025	10008167.0	22385416.0
32393583.0					
253	OCEAN CASINO RESORT	March	2025	8841515.0	25843086.0
34684601.0					
254	OCEAN CASINO RESORT	April	2025	5977368.0	24636951.0
30614319.0					
255	OCEAN CASINO RESORT	May	2025	12052625.0	30237289.0
42289914.0					
256	OCEAN CASINO RESORT	June	2025	14006551.0	29540710.0
43547261.0					
257	OCEAN CASINO RESORT	July	2025	10919112.0	32214930.0
43134042.0					
258	OCEAN CASINO RESORT	August	2025	15705864.0	35458777.0
51164641.0					
259	OCEAN CASINO RESORT	September	2025	13685844.0	25956968.0
39642812.0					
260	OCEAN CASINO RESORT	October	2025	12787062.0	26640630.0
39427692.0					
261	OCEAN CASINO RESORT	November	2025	11876986.0	27127226.0
39004212.0					

Preview - iGaming (Ocean Casino):

	casino	month	year	monthly_total_igaming_win
ytd_igaming_gross_revenue				
670	OCEAN CASINO RESORT	January	2022	1907261.0
1912202.0				
671	OCEAN CASINO RESORT	February	2022	1120750.0
3032973.0				
672	OCEAN CASINO RESORT	March	2022	2349866.0
5411346.0				
673	OCEAN CASINO RESORT	April	2022	1677837.0
7050220.0				
674	OCEAN CASINO RESORT	May	2022	2199468.0
9252045.0				
675	OCEAN CASINO RESORT	June	2022	2351737.0
11607208.0				
676	OCEAN CASINO RESORT	July	2022	2739278.0
14351377.0				
677	OCEAN CASINO RESORT	August	2022	2922331.0
17280286.0				
678	OCEAN CASINO RESORT	September	2022	2371149.0
NaN				

679	OCEAN CASINO RESORT	October	2022	3259173.0
				NaN

Preview - Sports Wagering (Ocean Casino):

	casino	month	year	monthly_internet_sports_win	monthly_retail_sports_win
1290	OCEAN CASINO RESORT	January	2022	229206.0	141925.0
1291	OCEAN CASINO RESORT	February	2022	293903.0	NaN
1292	OCEAN CASINO RESORT	March	2022	224942.0	NaN
1293	OCEAN CASINO RESORT	April	2022	65061.0	NaN
1294	OCEAN CASINO RESORT	May	2022	114270.0	17026.0
1295	OCEAN CASINO RESORT	June	2022	149023.0	7875.0
1296	OCEAN CASINO RESORT	July	2022	148990.0	49296.0
1297	OCEAN CASINO RESORT	August	2022	306333.0	152205.0
1298	OCEAN CASINO RESORT	September	2022	312312.0	224043.0
1299	OCEAN CASINO RESORT	October	2022	10880.0	71685.0

```
[17]: import pandas as pd
df = pd.read_csv(r"C:\Users\abidh\Documents\casino_revenue_clean.csv")
print(df['casino'].unique())
```

```
["BALLY'S ATLANTIC CITY" 'BORGATA HOTEL CASINO & SPA'
'CAESARS ATLANTIC CITY' 'GOLDEN NUGGET' 'HARD ROCK ATLANTIC CITY'
"HARRAH'S ATLANTIC CITY" "HARRAH'S REPORT ATLANTIC CITY"
"HARRAH'S RESORT ATLANTIC CITY" 'OCEAN CASINO RESORT'
'OCEAN CASINO RESORT AC' 'PREMIER ENT AC' 'PREMIER ENTERTAINMENT AC'
'RESORTS CASINO HOTEL' 'RESORTS CASINO HOTEL (DGMB CASINO'
'TROPICANA CASINO & RESORT' 'TROPICANA CASINO & RESORTS']
```

```
[19]: import pandas as pd

name_map = {
    'PREMIER ENTERTAINMENT AC': "BALLY'S ATLANTIC CITY",
    'PREMIER ENT AC': "BALLY'S ATLANTIC CITY",
    'OCEAN CASINO RESORT AC': 'OCEAN CASINO RESORT',
    'HARRAH'S RESORT ATLANTIC CITY': "HARRAH'S ATLANTIC CITY",
    'HARRAH'S REPORT ATLANTIC CITY': "HARRAH'S ATLANTIC CITY",
```

```

        'RESORTS CASINO HOTEL (DGMB CASINO)': 'RESORTS CASINO HOTEL',
        'TROPICANA CASINO & RESORTS': 'TROPICANA CASINO & RESORT',
    }

for filename in ['casino_revenue_clean', 'igaming_clean', 'sports_wagering_clean', 'all_casino_data']:
    path = rf"C:\Users\abidh\Documents\{filename}.csv"
    df = pd.read_csv(path)
    df['casino'] = df['casino'].replace(name_map)
    df.to_csv(path, index=False)
    print(f"Fixed: {filename}.csv")

df = pd.read_csv(r"C:\Users\abidh\Documents\casino_revenue_clean.csv")
ballys = df[df['casino'] == "BALLY'S ATLANTIC CITY"]
print(f"\nBally's records: {len(ballys)}")
print(ballys['year'].value_counts().sort_index())

```

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Fixed: casino_revenue_clean.csv
Fixed: igaming_clean.csv
Fixed: sports_wagering_clean.csv
Fixed: all_casino_data.csv

```

Bally's records: 48

```

year
2022    12
2023    12
2024    12
2025    12
Name: count, dtype: int64

```

```

[21]: import pandas as pd
df = pd.read_csv(r"C:\Users\abidh\Documents\casino_revenue_clean.csv")
print(df.groupby('casino')['year'].value_counts().unstack().fillna(0).
      astype(int))

```

year	2022	2023	2024	2025
casino				
BALLY'S ATLANTIC CITY	12	12	12	12
BORGATA HOTEL CASINO & SPA	12	12	12	12
CAESARS ATLANTIC CITY	12	12	12	12
GOLDEN NUGGET	12	12	12	12
HARD ROCK ATLANTIC CITY	12	12	12	12
HARRAH'S ATLANTIC CITY	12	12	12	12
OCEAN CASINO RESORT	12	12	12	12
RESORTS CASINO HOTEL	12	12	12	12
TROPICANA CASINO & RESORT	12	12	12	12

```
[22]: import pandas as pd
df = pd.read_csv(r"C:\Users\abidh\Documents\casino_revenue_clean.csv")
ballys = df[df['casino'].str.contains('BALLY', case=False, na=False)]
print(ballys[['casino', 'month', 'year', 'total_casino_win']].to_string())
```

	casino	month	year	total_casino_win
0	BALLY'S ATLANTIC CITY	January	2025	9668742
1	BALLY'S ATLANTIC CITY	February	2025	9224853
2	BALLY'S ATLANTIC CITY	March	2025	11041065
3	BALLY'S ATLANTIC CITY	April	2025	10273137
4	BALLY'S ATLANTIC CITY	May	2025	11865007
5	BALLY'S ATLANTIC CITY	June	2025	11089646
6	BALLY'S ATLANTIC CITY	July	2025	14119163
7	BALLY'S ATLANTIC CITY	August	2025	14451730
8	BALLY'S ATLANTIC CITY	September	2025	10127933
9	BALLY'S ATLANTIC CITY	October	2025	11334316
10	BALLY'S ATLANTIC CITY	November	2025	10679285
11	BALLY'S ATLANTIC CITY	December	2025	9150715
300	BALLY'S ATLANTIC CITY	January	2023	11544104
301	BALLY'S ATLANTIC CITY	February	2023	12269167
302	BALLY'S ATLANTIC CITY	March	2023	10550249
303	BALLY'S ATLANTIC CITY	April	2023	13793997
304	BALLY'S ATLANTIC CITY	May	2023	13197719
305	BALLY'S ATLANTIC CITY	June	2023	12431571
306	BALLY'S ATLANTIC CITY	July	2023	17203588
307	BALLY'S ATLANTIC CITY	August	2023	15009194
308	BALLY'S ATLANTIC CITY	September	2023	13140212
309	BALLY'S ATLANTIC CITY	October	2023	12436684
310	BALLY'S ATLANTIC CITY	November	2023	12084045
311	BALLY'S ATLANTIC CITY	December	2023	11517254
312	BALLY'S ATLANTIC CITY	January	2024	10070386
313	BALLY'S ATLANTIC CITY	February	2024	10657196
314	BALLY'S ATLANTIC CITY	March	2024	11733290
315	BALLY'S ATLANTIC CITY	April	2024	13119845
316	BALLY'S ATLANTIC CITY	May	2024	12425975
317	BALLY'S ATLANTIC CITY	June	2024	13555398
318	BALLY'S ATLANTIC CITY	July	2024	13304119
319	BALLY'S ATLANTIC CITY	August	2024	15638296
320	BALLY'S ATLANTIC CITY	September	2024	12568128
321	BALLY'S ATLANTIC CITY	October	2024	11472221
322	BALLY'S ATLANTIC CITY	November	2024	10838592
323	BALLY'S ATLANTIC CITY	December	2024	10008814
324	BALLY'S ATLANTIC CITY	January	2022	8783616
325	BALLY'S ATLANTIC CITY	February	2022	10736761
326	BALLY'S ATLANTIC CITY	March	2022	13478256
327	BALLY'S ATLANTIC CITY	April	2022	14418816
328	BALLY'S ATLANTIC CITY	May	2022	12670693

329	BALLY'S ATLANTIC CITY	June	2022	13547480
330	BALLY'S ATLANTIC CITY	July	2022	16820296
331	BALLY'S ATLANTIC CITY	August	2022	15735889
332	BALLY'S ATLANTIC CITY	September	2022	15892155
333	BALLY'S ATLANTIC CITY	October	2022	10828239
334	BALLY'S ATLANTIC CITY	November	2022	10034509
335	BALLY'S ATLANTIC CITY	December	2022	10954715

```
[23]: import pandas as pd

for filename in ['casino_revenue_clean', 'igaming_clean', 'sports_wagering_clean', 'all_casino_data']:
    path = rf"C:\Users\abidh\Documents\{filename}.csv"
    df = pd.read_csv(path)
    df['date'] = pd.to_datetime(df['month'] + ' ' + df['year']).astype(str),
    format='%B %Y', errors='coerce')
    df = df.sort_values(['casino', 'date']).reset_index(drop=True)
    df.to_csv(path, index=False)
    print(f"Sorted: {filename}.csv")

print("Done! All files now sorted by casino then date.")
```

```
Sorted: casino_revenue_clean.csv
Sorted: igaming_clean.csv
Sorted: sports_wagering_clean.csv
Sorted: all_casino_data.csv
Done! All files now sorted by casino then date.
```

```
[25]: import pandas as pd

igaming_name_map = {
    'Premier Entertainment AC': "BALLY'S ATLANTIC CITY",
    'RESORTS DIGITAL GAMING': 'RESORTS CASINO HOTEL',
    'TROPICANA CASINO AND RESORT': 'TROPICANA CASINO & RESORT',
    'TROPICANA CASINO AND RESORT (ONLINE)': 'TROPICANA CASINO & RESORT',
    'CAESARS INTERACTIVE ENTERTAINMENT': 'CAESARS ATLANTIC CITY',
    '(AFFILIATE OF HARRAH'S ATLANTIC CITY)': "HARRAH'S ATLANTIC CITY",
    '(AFFILIATE OF TROPICANA CASINO AND RESORT)': 'TROPICANA CASINO & RESORT',
    'Golden Nugget Online Gaming': 'GOLDEN NUGGET',
    'GOLDEN NUGGET ATLANTIC CITY': 'GOLDEN NUGGET',
}

for filename in ['igaming_clean', 'all_casino_data']:
    path = rf"C:\Users\abidh\Documents\{filename}.csv"
    df = pd.read_csv(path)
    df['casino'] = df['casino'].replace(igaming_name_map)
    df.to_csv(path, index=False)
```

```

print(f"Fixed: {filename}.csv")

# Verify
df = pd.read_csv(r"C:\Users\abidh\Documents\igaming_clean.csv")
print("\nRecords per casino per year after fix:")
igaming_only = df[df['data_type'] == 'igaming']
print(igaming_only.groupby('casino')['year'].value_counts().unstack().fillna(0).
      astype(int))

```

Fixed: igaming_clean.csv
Fixed: all_casino_data.csv

Records per casino per year after fix:

year	2022	2023	2024	2025
casino				
BALLY'S ATLANTIC CITY	12	12	12	12
BORGATA HOTEL CASINO & SPA	12	12	12	12
CAESARS ATLANTIC CITY	12	12	6	0
GOLDEN NUGGET	13	12	12	12
HARD ROCK ATLANTIC CITY	12	12	12	12
HARRAH'S ATLANTIC CITY	0	3	12	12
OCEAN CASINO RESORT	12	12	12	12
RESORTS CASINO HOTEL	12	12	12	12
TROPICANA CASINO & RESORT	12	12	12	12

```

[26]: import pandas as pd

path = r"C:\Users\abidh\Documents\igaming_clean.csv"
df = pd.read_csv(path)

# Check the duplicate
gn_2022 = df[(df['casino'] == 'GOLDEN NUGGET') & (df['year'] == 2022)]
print(gn_2022[['casino', 'month', 'year', 'monthly_total_igaming_win']])

```

	casino	month	year	monthly_total_igaming_win
149	GOLDEN NUGGET	June	2022	35705464.0
150	GOLDEN NUGGET	July	2022	34529509.0
151	GOLDEN NUGGET	August	2022	31376905.0
152	GOLDEN NUGGET	September	2022	34167048.0
153	GOLDEN NUGGET	October	2022	38383650.0
154	GOLDEN NUGGET	November	2022	37710941.0
155	GOLDEN NUGGET	December	2022	37786899.0
192	GOLDEN NUGGET	NaN	2022	29693191.0
193	GOLDEN NUGGET	January	2022	36485877.0
194	GOLDEN NUGGET	February	2022	34560269.0
195	GOLDEN NUGGET	March	2022	38280684.0
196	GOLDEN NUGGET	April	2022	38142387.0
197	GOLDEN NUGGET	NaN	2022	4910572.0

```
[27]: import pandas as pd

path = r"C:\Users\abidh\Documents\igaming_clean.csv"
df = pd.read_csv(path)

# Remove rows with missing month
before = len(df)
df = df.dropna(subset=['month'])
after = len(df)
print(f"Removed {before - after} rows with missing months")

df.to_csv(path, index=False)

# Verify Golden Nugget now has 12 records in 2022
gn = df[(df['casino'] == 'GOLDEN NUGGET') & (df['year'] == 2022)]
print(f"Golden Nugget 2022 records: {len(gn)}")
print(df.groupby('casino')['year'].value_counts().unstack().fillna(0).
      .astype(int))
```

Removed 2 rows with missing months

Golden Nugget 2022 records: 11

year	2022	2023	2024	2025
casino				
BALLY'S ATLANTIC CITY	12	12	12	12
BORGATA HOTEL CASINO & SPA	12	12	12	12
CAESARS ATLANTIC CITY	12	12	6	0
GOLDEN NUGGET	11	12	12	12
HARD ROCK ATLANTIC CITY	12	12	12	12
HARRAH'S ATLANTIC CITY	0	3	12	12
OCEAN CASINO RESORT	12	12	12	12
RESORTS CASINO HOTEL	12	12	12	12
TROPICANA CASINO & RESORT	12	12	12	12

```
[28]: import pandas as pd

df = pd.read_csv(r"C:\Users\abidh\Documents\igaming_clean.csv")

gn_2022 = df[(df['casino'] == 'GOLDEN NUGGET') & (df['year'] == 2022)]
print(gn_2022[['casino', 'month', 'year', 'monthly_total_igaming_win']].
      .to_string())
```

	casino	month	year	monthly_total_igaming_win
149	GOLDEN NUGGET	June	2022	35705464.0
150	GOLDEN NUGGET	July	2022	34529509.0
151	GOLDEN NUGGET	August	2022	31376905.0
152	GOLDEN NUGGET	September	2022	34167048.0
153	GOLDEN NUGGET	October	2022	38383650.0
154	GOLDEN NUGGET	November	2022	37710941.0

155	GOLDEN NUGGET	December	2022	37786899.0
192	GOLDEN NUGGET	January	2022	36485877.0
193	GOLDEN NUGGET	February	2022	34560269.0
194	GOLDEN NUGGET	March	2022	38280684.0
195	GOLDEN NUGGET	April	2022	38142387.0

```
[29]: import pandas as pd

df = pd.read_csv(r"C:\Users\abidh\Documents\igaming_clean.csv")

print(df.groupby(['casino', 'year'])['month'].count().unstack().fillna(0).
      <math>\rightarrow</math>astype(int))
```

year	2022	2023	2024	2025
casino				
BALLY'S ATLANTIC CITY	12	12	12	12
BORGATA HOTEL CASINO & SPA	12	12	12	12
CAESARS ATLANTIC CITY	12	12	6	0
GOLDEN NUGGET	11	12	12	12
HARD ROCK ATLANTIC CITY	12	12	12	12
HARRAH'S ATLANTIC CITY	0	3	12	12
OCEAN CASINO RESORT	12	12	12	12
RESORTS CASINO HOTEL	12	12	12	12
TROPICANA CASINO & RESORT	12	12	12	12

```
[30]: import pandas as pd

df = pd.read_csv(r"C:\Users\abidh\Documents\casino_revenue_clean.csv")
casino_df = df[['casino', 'month', 'year', 'table_games_win', 'slot_win',
               <math>\rightarrow</math>'total_casino_win']]
casino_df.to_csv(r"C:\Users\abidh\Documents\casino_revenue_final.csv",
               <math>\rightarrow</math>index=False)
print("Casino Revenue:")
print(casino_df.head())
```

Casino Revenue:

	casino	month	year	table_games_win	slot_win \
0	BALLY'S ATLANTIC CITY	January	2022	3045482	5738134
1	BALLY'S ATLANTIC CITY	February	2022	3115899	7620862
2	BALLY'S ATLANTIC CITY	March	2022	5783567	7694689
3	BALLY'S ATLANTIC CITY	April	2022	5003408	9415408
4	BALLY'S ATLANTIC CITY	May	2022	2351636	10319057

	total_casino_win
0	8783616
1	10736761
2	13478256
3	14418816

```
[31]: df = pd.read_csv(r"C:\Users\abidh\Documents\igaming_clean.csv")
igaming_df = df[['casino', 'month', 'year', 'monthly_total_igaming_win']]
igaming_df.to_csv(r"C:\Users\abidh\Documents\igaming_final.csv", index=False)
print("\niGaming:")
print(igaming_df.head())
```

iGaming:

	casino	month	year	monthly_total_igaming_win
0	HARRAH'S ATLANTIC CITY	September	2024	1574804.0
1	HARRAH'S ATLANTIC CITY	October	2024	1848249.0
2	HARRAH'S ATLANTIC CITY	November	2024	1377376.0
3	HARRAH'S ATLANTIC CITY	December	2024	1444650.0
4	HARRAH'S ATLANTIC CITY	January	2025	1276401.0

```
[32]: df = pd.read_csv(r"C:\Users\abidh\Documents\sports_wagering_clean.csv")
sports_df = df[['casino', 'month', 'year', 'monthly_retail_sports_win',
               ↪ 'monthly_internet_sports_win']]
sports_df['total_sports_win'] = sports_df['monthly_retail_sports_win'].
               ↪ fillna(0) + sports_df['monthly_internet_sports_win'].fillna(0)
sports_df.to_csv(r"C:\Users\abidh\Documents\sports_wagering_final.csv",
               ↪ index=False)
print("\nSports Wagering:")
print(sports_df.head())
```

Sports Wagering:

	casino	month	year	monthly_retail_sports_win	monthly_internet_sports_win	total_sports_win
0	(AFFILIATE OF HARRAH'S ATLANTIC CITY)	September	2024	NaN	54266.0	54266.0
1	(AFFILIATE OF HARRAH'S ATLANTIC CITY)	October	2024	NaN	128692.0	128692.0
2	(AFFILIATE OF HARRAH'S ATLANTIC CITY)	November	2024	NaN	115995.0	115995.0
3	(AFFILIATE OF HARRAH'S ATLANTIC CITY)	December	2024	NaN	171537.0	171537.0
4	(AFFILIATE OF HARRAH'S ATLANTIC CITY)	January	2025	NaN	297320.0	297320.0

C:\Users\abidh\AppData\Local\Temp\ipykernel_30140\180203620.py:3:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
sports_df['total_sports_win'] =  
sports_df['monthly_retail_sports_win'].fillna(0) +  
sports_df['monthly_internet_sports_win'].fillna(0)
```

```
[33]: import pandas as pd  
  
# CASINO REVENUE  
df = pd.read_csv(r"C:\Users\abidh\Documents\casino_revenue_clean.csv")  
casino_df = df[['casino', 'month', 'year', 'table_games_win', 'slot_win',  
               ↪ 'total_casino_win']].copy()  
casino_df.to_csv(r"C:\Users\abidh\Documents\casino_revenue_final.csv",  
               ↪ index=False)  
print(f"Casino Revenue: {len(casino_df)} records")  
print(casino_df['casino'].unique())  
  
# IGAMING  
df = pd.read_csv(r"C:\Users\abidh\Documents\igaming_clean.csv")  
igaming_df = df[['casino', 'month', 'year', 'monthly_total_igaming_win']].copy()  
  
# Remove affiliates and non-casino entities  
exclude = ['AFFILIATE', 'INTERACTIVE', 'DIGITAL', 'ONLINE GAMING']  
mask = ~igaming_df['casino'].str.upper().str.contains(''.join(exclude),  
               ↪ na=False)  
igaming_df = igaming_df[mask]  
igaming_df.to_csv(r"C:\Users\abidh\Documents\igaming_final.csv", index=False)  
print(f"niGaming: {len(igaming_df)} records")  
print(igaming_df['casino'].unique())  
  
# SPORTS WAGERING  
df = pd.read_csv(r"C:\Users\abidh\Documents\sports_wagering_clean.csv")  
sports_df = df[['casino', 'month', 'year', 'monthly_retail_sports_win',  
               ↪ 'monthly_internet_sports_win']].copy()  
  
# Remove affiliates and racetracks  
exclude_sw = ['AFFILIATE', 'MEADOWLANDS', 'MONMOUTH', 'FREEHOLD', 'DARBY',  
              'FR PARK', 'BOARDWALK REGENCY', 'DIGITAL', 'OPERATING', 'ONLINE']  
mask = ~sports_df['casino'].str.upper().str.contains(''.join(exclude_sw),  
               ↪ na=False)  
sports_df = sports_df[mask]  
  
# Standardize names  
sw_name_map = {  
    'HARD ROCK HOTEL & CASINO ATLANTIC CITY': 'HARD ROCK ATLANTIC CITY',  
    'HARD ROCK HOTEL AND CASINO ATLANTIC CITY': 'HARD ROCK ATLANTIC CITY',  
    'Borgata Hotel Casino & Spa': 'BORGATA HOTEL CASINO & SPA',
```

```

    'TROPICANA CASINO AND RESORT': 'TROPICANA CASINO & RESORT',
    'Tropicana Casino & Resort': 'TROPICANA CASINO & RESORT',
    'Tropicana Casino & Resort (Online)': 'TROPICANA CASINO & RESORT',
    'GOLDEN NUGGET ONLINE GAMING': 'GOLDEN NUGGET',
    'HARRAH'S ATLANTIC CITY OPERATING': 'HARRAH'S ATLANTIC CITY',
    'Darby Development (Monmouth Park)': None,
}
sports_df['casino'] = sports_df['casino'].replace(sw_name_map)

# Create total sports win column
sports_df['total_sports_win'] = sports_df['monthly_retail_sports_win'].
    ↪ fillna(0) + sports_df['monthly_internet_sports_win'].fillna(0)
sports_df.to_csv(r"C:\Users\abidh\Documents\sports_wagering_final.csv",
    ↪ index=False)
print(f"\nSports Wagering: {len(sports_df)} records")
print(sports_df['casino'].unique())

```

Casino Revenue: 432 records

```

["BALLY'S ATLANTIC CITY" 'BORGATA HOTEL CASINO & SPA'
 'CAESARS ATLANTIC CITY' 'GOLDEN NUGGET' 'HARD ROCK ATLANTIC CITY'
 "HARRAH'S ATLANTIC CITY" 'OCEAN CASINO RESORT' 'RESORTS CASINO HOTEL'
 'TROPICANA CASINO & RESORT']

```

iGaming: 392 records

```

["HARRAH'S ATLANTIC CITY" 'TROPICANA CASINO & RESORT'
 "BALLY'S ATLANTIC CITY" 'BORGATA HOTEL CASINO & SPA'
 'CAESARS ATLANTIC CITY' 'GOLDEN NUGGET' 'HARD ROCK ATLANTIC CITY'
 'OCEAN CASINO RESORT' 'RESORTS CASINO HOTEL']

```

Sports Wagering: 456 records

```

["BALLY'S ATLANTIC CITY" 'BORGATA HOTEL CASINO & SPA'
 'CAESARS ATLANTIC CITY' 'GOLDEN NUGGET' 'HARD ROCK ATLANTIC CITY'
 "HARRAH'S ATLANTIC CITY" 'OCEAN CASINO RESORT' 'RESORTS CASINO HOTEL'
 'TROPICANA CASINO & RESORT']

```

```

[34]: import pandas as pd

print("Casino Revenue Final columns:")
df = pd.read_csv(r"C:\Users\abidh\Documents\casino_revenue_final.csv")
print(df.columns.tolist())

print("\niGaming Final columns:")
df = pd.read_csv(r"C:\Users\abidh\Documents\igaming_final.csv")
print(df.columns.tolist())

print("\nSports Wagering Final columns:")
df = pd.read_csv(r"C:\Users\abidh\Documents\sports_wagering_final.csv")

```

```
print(df.columns.tolist())
```

Casino Revenue Final columns:

```
['casino', 'month', 'year', 'table_games_win', 'slot_win', 'total_casino_win']
```

iGaming Final columns:

```
['casino', 'month', 'year', 'monthly_total_igaming_win']
```

Sports Wagering Final columns:

```
['casino', 'month', 'year', 'monthly_retail_sports_win',  
'monthly_internet_sports_win', 'total_sports_win']
```

```
[ ]:
```